Statistics

Definition –

A branch of applied maths that involves the collection description, presentation, analysis and Intereprtation of numerical data. is called statistics.

☆ Type of Data –

① structured data

② Unstructured data

// Stages of stats –

① collection of data

② Organizing of data

③ Presentation of data

④ Analysis of data

⑤ Intereprefation of data

☆ Type of stats -

① Descriptive stats

Available data sample or population on it we perform action like analysing. discribe, summerize. it called descriptive stats.

② Inferential stats -

On describe data we perform interepetation like hypothesis testing on the data for example Z-test, T-test, F-test, chi-squere test. is called inferential statistics

① Descriptive stats.

① Univariate Des. stats.

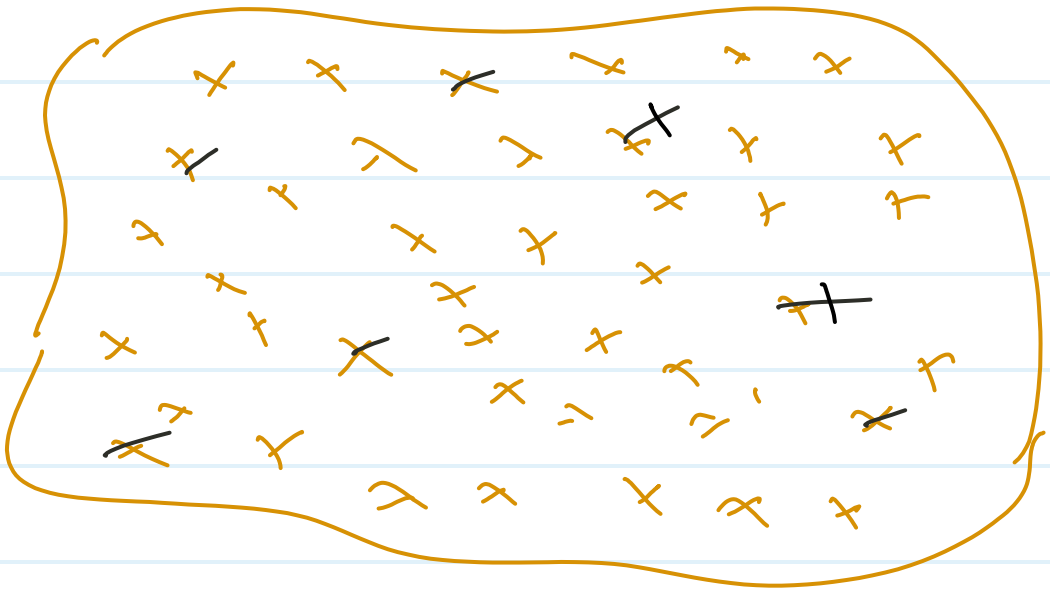
② Bivariate Des. stats.

③ Multi variate Des. stats.

✡ Population & Sample

Entire data available is call population.

Population = N



Sample = n

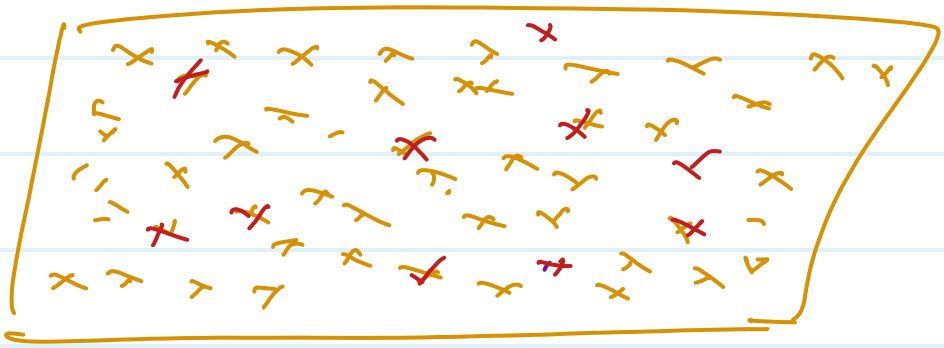
✡ Sampling methods –

// Type of Sampling method

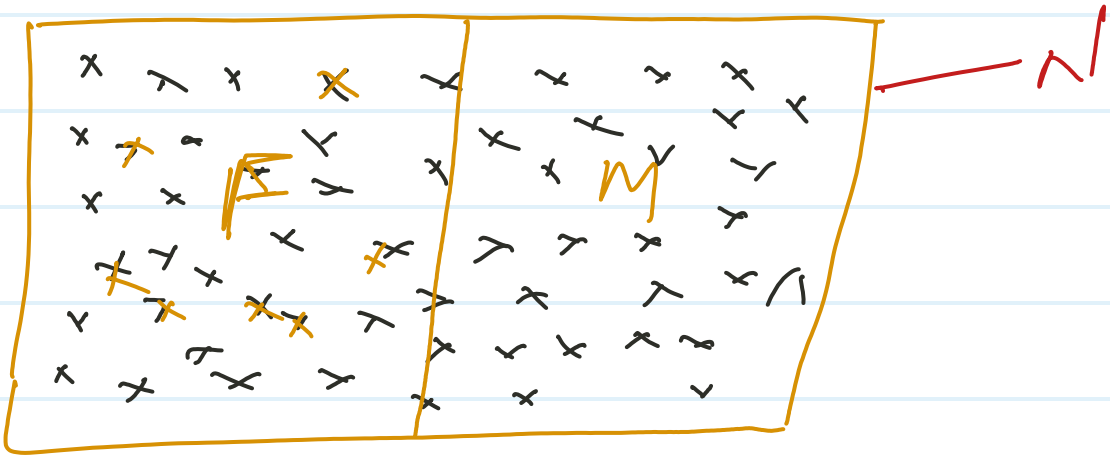
1. Probability Sampling
2. Non-probability sampling.

Probability sampling

① Simple random Sampling.-



② stratified Sampling



> 18

Gender.

③ Systematic sampling.-

n^{th} select

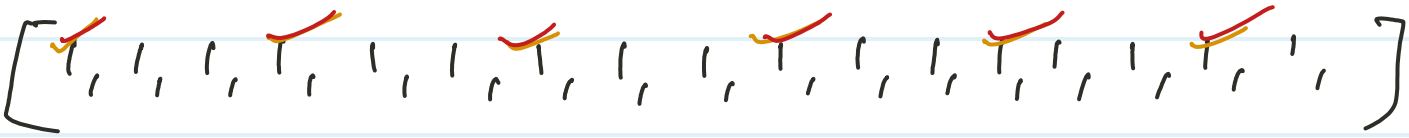
population - 500
sample - 50

$$= \frac{50\cancel{0}}{5\cancel{0}}$$

$$= 10$$

> 18

10th, 20th, 30th



② Non-probability Sampling -

① Convinience Sampling -

Research -

How to calculate sample size?

* Cochran Formula

$$n_0 = \frac{Z^2 pq}{e^2}$$ – infinite population

$$n = \frac{n_0}{1 + \frac{(n_0 - 1)}{N}}$$ – Required population.

e = margin of error 5%

p = Population proportion 50%

q = $1 - p$

Z = value from Z-table

C.I.		α
90 %	-	0.1
95 %	-	0.05
99 %	-	0.01

0.975

<u>Z-table</u>

$$\alpha = 0.05$$

$$1 - 0.05 = 0.95$$

$$Z = 1.6 + 0.05$$

$$\boxed{Z = 1.65}$$

$$= \frac{(1.65)^2 \times 0.5 \times (1 - 0.5)}{(0.05)^2}$$

$$= \frac{2.72 \times 0.5 \times 0.5}{0.025}$$

$$n_0 = \frac{0.68}{0.025} = \underline{\underline{27.2}}$$

Required population $N = 10000$

$$n = \frac{n_0}{1 + \frac{(n_0 - 1)}{N}}$$

$$= \frac{27.2}{1 + \frac{(27.2 - 1)}{10000}}$$

$$\Rightarrow 27.12$$

$$28 / 27$$

If population = 10000

Sample = 28

Descriptive stats -

1. measure of Center tendency
2. measure of Dispersion or variation
3. measure of position
4. measure of shape

☆ Measure of Center Tendency

1. mean -

$[2, 3, 4, 5, 6]$

$$\text{mean} = \frac{2+3+4+5+6}{5} = \frac{20}{5}$$

$$= 4$$

population mean $= \mu$

Sample mean $= \bar{x}$

② median –

I-case [6, 2, 4, 5, 3]

data sort

[2, 3, 4, 5, 6]

↑

median = 4

II-Case

[2, 3, 5, 7, 8, 9, 11, 13]

$$\text{median} = \frac{7+8}{2} \Rightarrow 7.5$$

③ mode –

(i) uni-modal
(ii) Bi-modal
(iii) multi-modal

① [2, 3, 4, 5, 5, 6]

mode = 5

② [2, 2, 4, 5, 6, 6]

mode = 2, 6

③ [2, 2, 4, 4, 5, 5, 6, 7, 8]

mode = 2, 4, 5

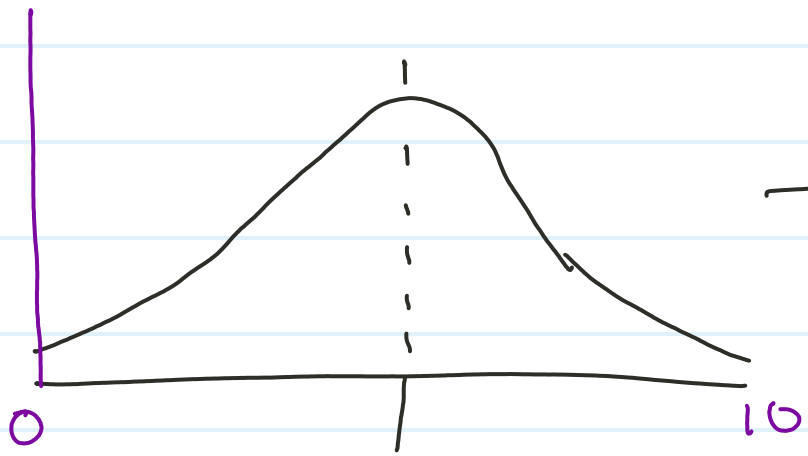
→ main application

→ outlier handeling

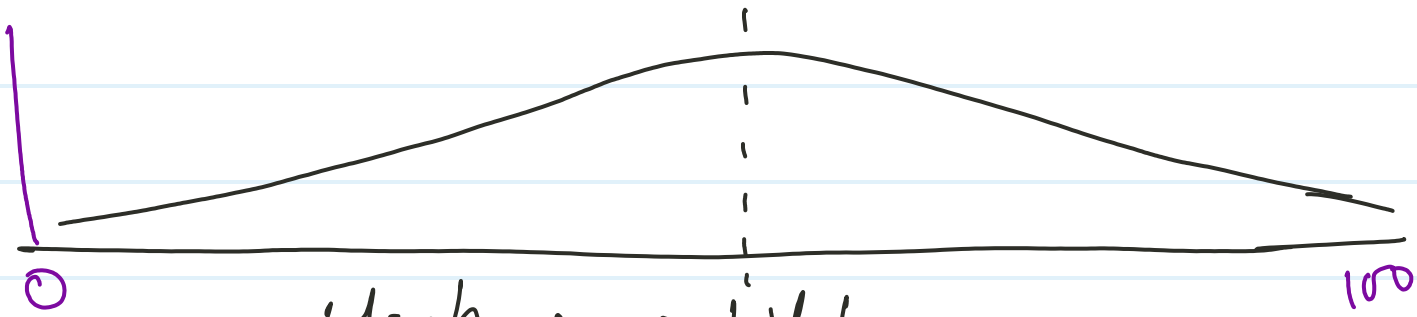
→ missing value handel

	age	salary	Result
	20	10	P
	18	15	F
Null	—	—	P
	15	30	F
	22	—	—
Null	—	35	—
	27	120	F

* measure of Dispersion or variance



— low variability



High variability

① mean absolute deviation -

The mean absolute devi. of a dataset is the avg. distance b/w each data point and the mean.

$$\Rightarrow \quad \frac{1}{N} \sum_{i=1}^{n} |x_i - \bar{x}|$$

Data - [10, 15, 15, 17, 18, 21]

$$\text{mean} = \frac{96}{6} = 16$$

$$|10-16| = |-6| = 6$$
$$|15-16| = |-1| = 1$$
$$|15-16| = 1$$
$$|17-16| = 1$$
$$|18-16| = 2$$
$$|21-16| = 5$$
$$16$$

$$\Rightarrow \frac{16}{6} \Rightarrow 2.67$$

② Variance -

It tells the degree of spread in dataset.

High variability - datapoint spread widely

low variability - data point close to mean.

population $\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (X_i - \mu)^2$

sample $s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{x})^2$

*Note - $n-1$ is a degree of freedom

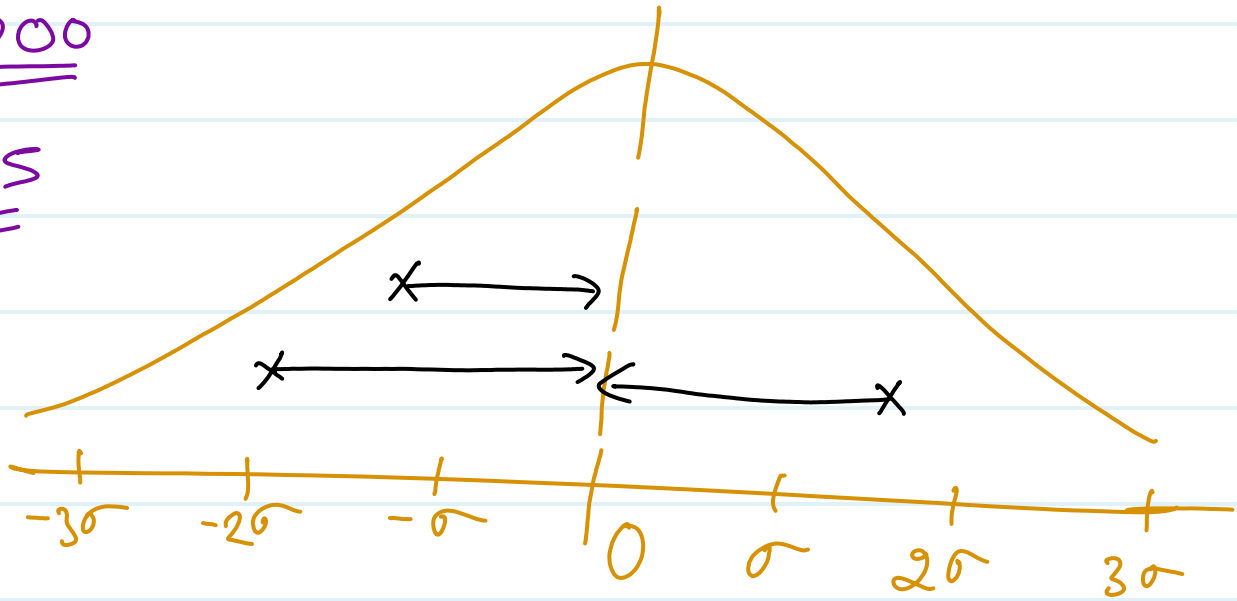
OR

Besils correction

For keep away result from biased.

200

95



③ standard deviation -

The square root of variance is called st. dev..

The farthes the data points from the higher the deviation

Pop $\sigma = \sqrt{\frac{1}{N} \sum_{i=1} (x_i - \mu)^2}$

Sample $S = \sqrt{\frac{1}{n-1} \sum_{i=1} (x_i - \bar{x})^2}$

③ Range -

[1, 2, 5, 6, 11, 15, 19, 25, 30]

max - 30

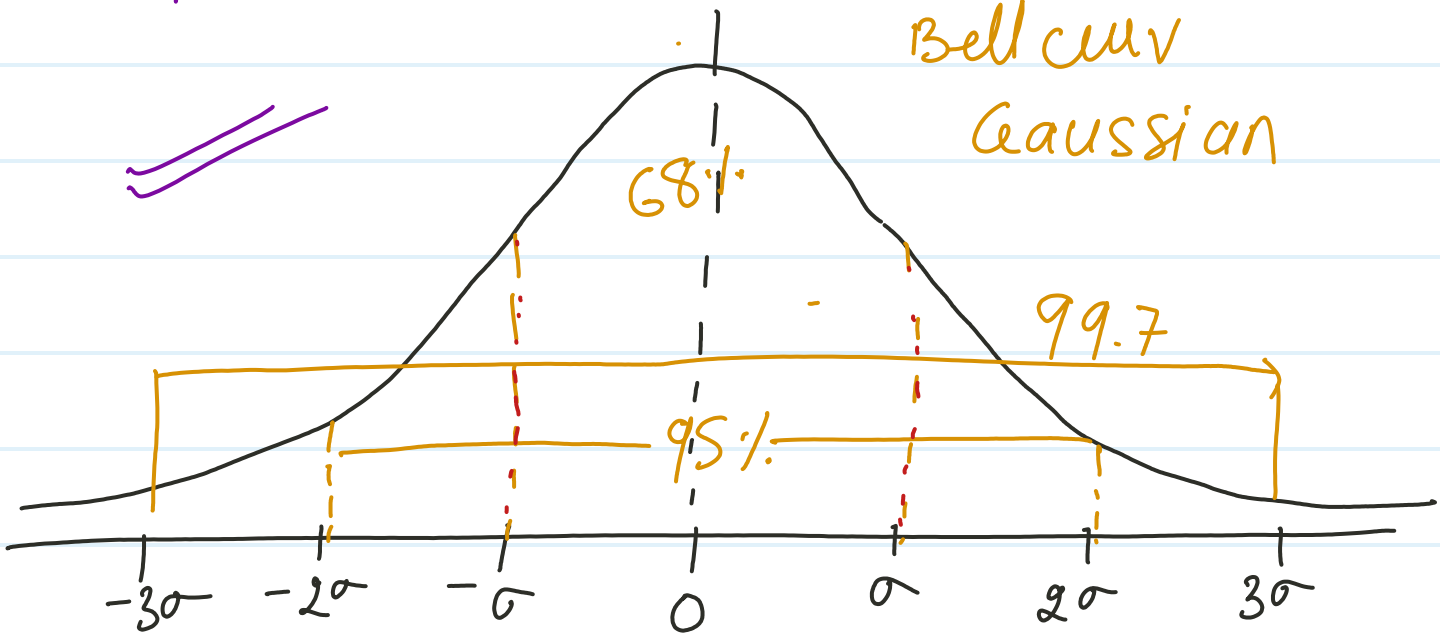
min - 1

range = max - min

= 30 - 1 = 29

* Emperical Rule

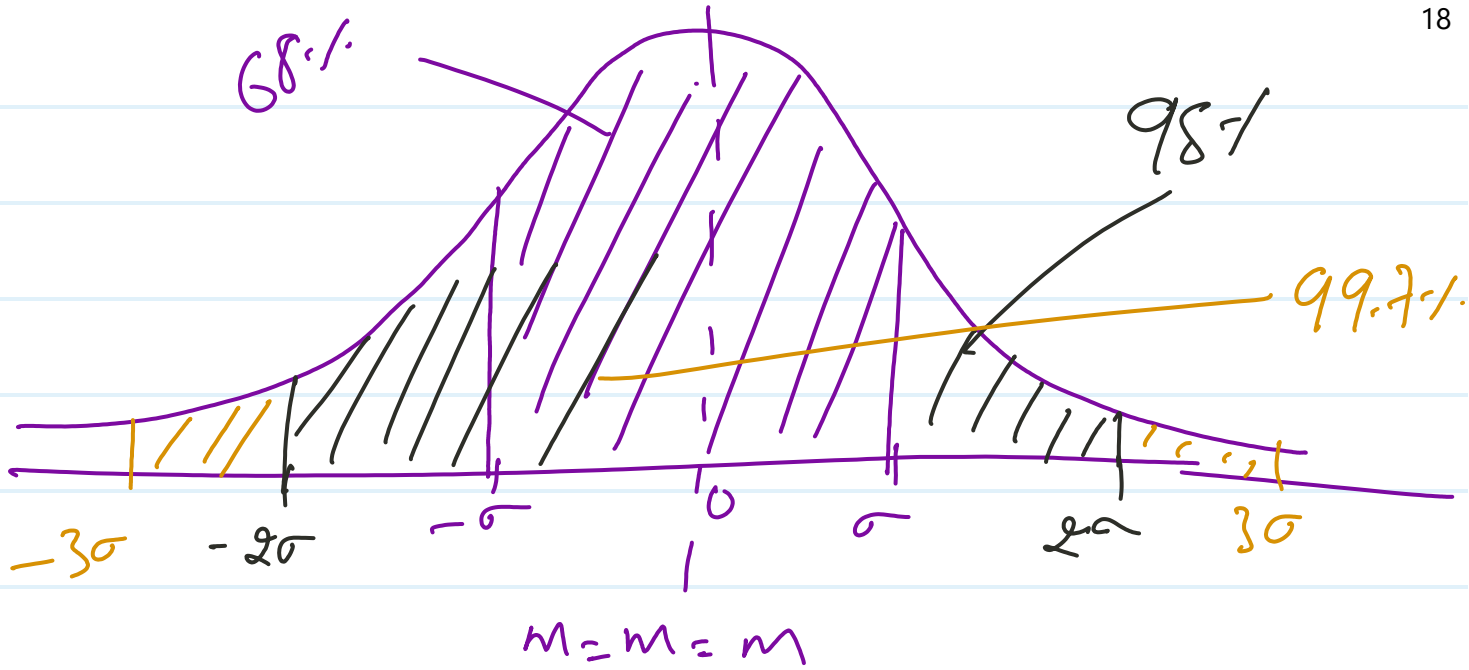
Normal
Bell curv
Gaussian



Normal Distribution Curv

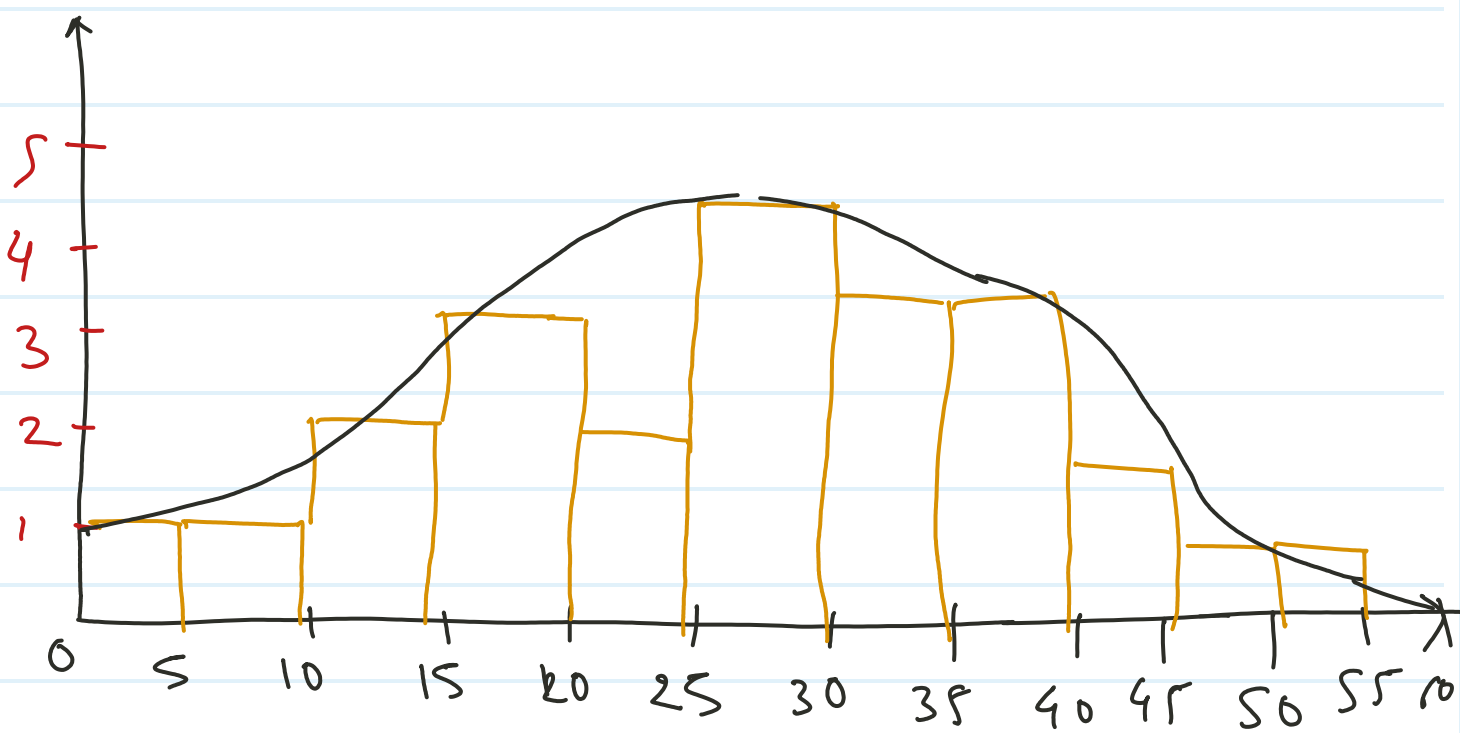
68 — 95 — 99.7

σ 2σ 3σ



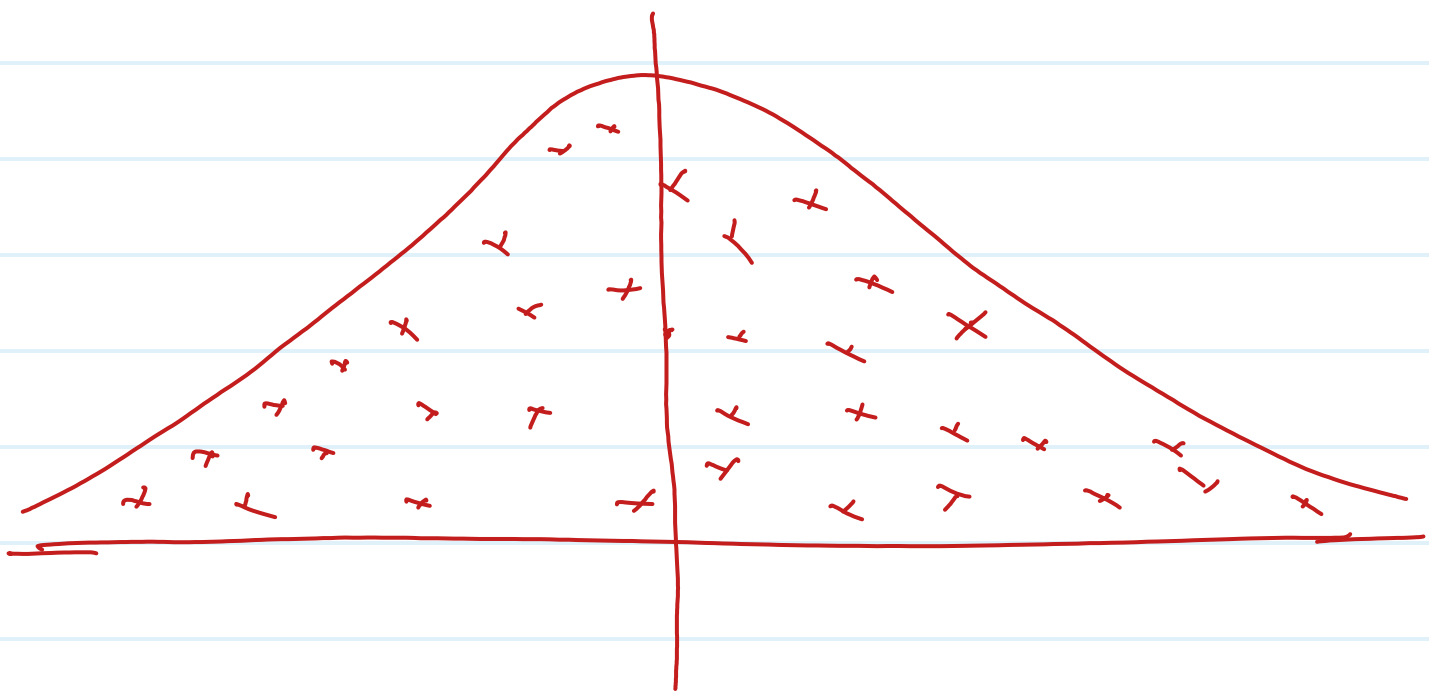
[3, 7, 11, 13, 16, 18, 19, 21, 23, 26, 27, 28, 29, 31, 32, 34, 36, 37, 39, 41, 44, 47, 48, 52, 57]

Range = 57 − 3 = 54/5 ⇒ 10.8



Data analysis ✓

Data analytics ✓



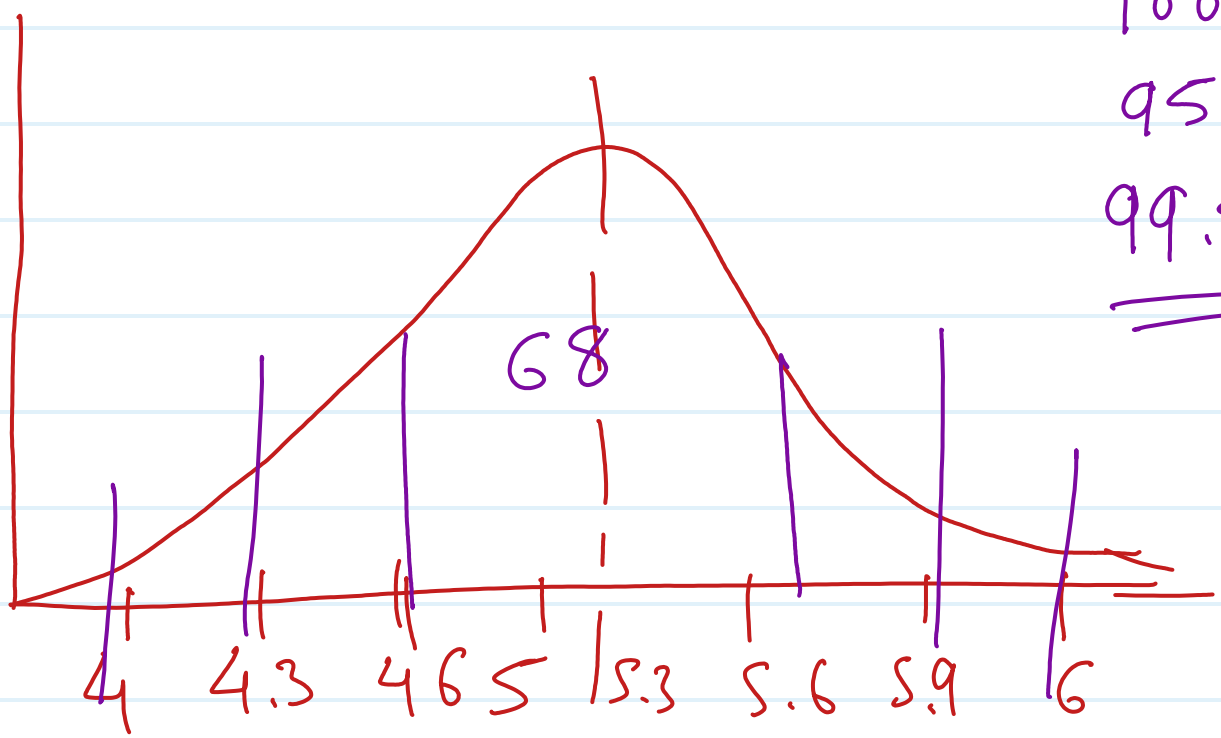
M = M = M

50 - 25 = 25 25

50 - 15 = 35

adult > 18

10k = height



100
95
99.5

$\boxed{68 \quad \sigma}$ — $\boxed{95 \quad 2\sigma}$ — $\boxed{99.7 \quad 3\sigma}$

$$[10, 12, 15, 9, 17, 20, 25, 7, 8]$$

x_i	$\bar{x}$	$(x_i - \bar{x})$
10	13.6	$10 - 13.6 = -3.6$
12		$12 - 13.6 = -1.6$
15		$15 - 13.6 = 1.4$
9		$9 - 13.6 = -4.6$
17		$17 - 13.6 = 3.4$
20		$20 - 13.6 = 6.4$
25		$25 - 13.6 = 11.4$
7		$7 - 13.6 = -6.6$
8		$8 - 13.6 = -5.6$

$$\Rightarrow \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$\Rightarrow \frac{1}{9-1} \times \left[-3.6 + (-1.6) + 1.4 + (-4.6) + 3.4 + 6.4 + 11.4 + (-6.6) + (-5.6) \right]$$

$$\Rightarrow \frac{1}{8} \times 0.7$$

$$\left[-3.6 - 1.6 + 1.4 - 4.6 + 3.4 + 6.4 + 11.4 - 6.6 - 5.6\right]$$

$$\Rightarrow \frac{0.6}{8} \qquad \frac{9.8}{8}$$

$$\underline{0.75} \qquad \underline{1.22}$$

$$\Rightarrow \sqrt{\frac{1}{8} \times \frac{0.6}{8}}$$

$$\Rightarrow \sqrt{0.75} \Rightarrow \underline{\underline{0.86}}$$